

HICHAM KRAOU

Full Stack & AI Developer

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PROFESSIONAL SUMMARY

Full Stack & AI Developer with proven experience delivering production-grade SaaS platforms, AI-powered automation systems, and multi-agent architectures. Proven track record designing and deploying real-world OpenClaw agent systems – including multi-agent orchestration, AI-driven web scraping pipelines, and production database integration with Firecrawl and MySQL. Specialized in building scalable applications with Next.js/Node.js, multi-tenant SaaS, conversational AI, Docker-based deployments, and LLM-oriented architectures including RAG, fine-tuning, and tool-using autonomous agents.

PROJECTS

OpenClaw AI Agent Platform – Multi-Agent Orchestration & Real-Estate Scraping Pipeline

Self-Hosted AI Gateway (Docker)

May 2026

Architected a multi-agent AI system on OpenClaw – the open-source, self-hosted AI agent framework (MIT, 250K+ GitHub stars). Deployed via Docker with a gateway proxying to OpenRouter LLMs, Firecrawl for web intelligence, and MySQL for structured data persistence.

- Designed and deployed a two-tier agent architecture: an Orchestrator agent that routes user requests by intent, and a dedicated RE-Scraper subagent that autonomously scrapes real-estate websites using Firecrawl, extracting structured rental unit data via AI reasoning into a 33-column MySQL schema with UPSERT deduplication
- Built the RE-Scraper agent with full autonomy: it reads raw web content (no regex/selectors), analyzes page structure, decides strategy (pagination, detail pages, hidden APIs, PDFs), extracts 33 normalized fields per unit, validates with sanity checks, and upserts into MySQL with checkpoint/resume capability for long-running scrapes
- Produced a working result: discovered a hidden JSON API on a real-estate site, scraped 262 rental units across 20 floors, extracted 4 typologies with availability statuses, and stored all data with 100% integrity into the database
- Implemented production-grade features: delta scraping (new/changed/unchanged detection), structured reporting (JSON + human-readable), French-Canadian real-estate field normalization (boolean, numeric, typology mappings), and unit conversion (sq ft → m²)
- Configured Firecrawl plugin for web search + scraping, OpenRouter for LLM routing, Tailscale-aware gateway networking, session memory, and OpenTelemetry diagnostics
- Stack: OpenClaw, Docker, Firecrawl API, MySQL, OpenRouter (owl-alpha), Python/uv, Linux

Qrox – Multi-Model AI Chat Platform

Live Demo: qrox.vercel.app

Feb 2026

Perplexity-inspired chat UI enabling users to interact with 500+ frontier AI models (OpenAI, Claude, Gemini, Mistral, xAI) with zero API key management through innovative user-pays architecture.

- Architected dual-authentication system (JWT + Puter SDK guest mode) achieving <10-second session initialization for 99% of users with automatic fallback and race condition handling

- Engineered real-time streaming chat with markdown rendering, syntax highlighting, multi-image vision support (10 concurrent uploads), and native AI image generation via function calling

- Implemented atomic conversation persistence with balance guards preventing ghost records during API failures, achieving 99.9% data integrity across 1,000+ test conversations

- Deployed production-ready application on Vercel with PostgreSQL backend, achieving <200ms database latency and full mobile responsiveness

- Tech: Next.js 16, Puter.js AI API, PostgreSQL, Prisma, TailwindCSS, shadcn/ui, TypeScript

HRAI – AI-Powered HR Assistant

Confidential Client, Rabat

Jan – Feb 2026

HRAI is an intelligent AI-powered HR assistant that enables non-technical staff to query a confidential personnel database using natural language (French). It replaces a manual process where HR had to call IT for every data request, with an autonomous conversational interface that generates and executes SQL from plain French questions.

- Architected & led a complete v3.0 → v4.0 transformation across 4 epics (53 functional requirements, ~405h effort)

- Designed the Semantic Tool Contract – a novel pattern where MCP tools return explicit semantic states (NO_DATA_FOUND, SUCCESS_WITH_DATA, ERROR) plus conversation_mode and next_action guidance, transforming tools from passive data sources into active dialogue partners

- Implemented fuzzy entity matching using Sentence Transformers (paraphrase-multilingual-MiniLM-L12-v2) with a 3-tier confidence system: ≥0.70 auto-match, 0.40-0.69 Differential Diagnosis, <0.40 ask user to rephrase

- Created the LEGO Assembly pattern for Text-to-SQL – pre-verified JOIN blocks from mdl.json (trusted structure) + LLM-generated WHERE clauses (flexible intent), solving the reliability-flexibility paradox in NL-to-SQL systems

- Built a self-healing SQL validator inspired by Rust compiler errors – provides exact error location, closest fuzzy column matches, and one-attempt auto-correction policy

- Refactored monolithic Docker deployment to isolated MCP microservices (8 containers), enabling parallel development and independent scaling

- Established CI/CD for knowledge base – syntax, schema, SQL, and circular dependency validation on every mdl.json change

- Implemented Differential Diagnosis menus – adapted from medical diagnosis methodology, presents top-3 options when queries are ambiguous

- Designed Query Relaxation Cascade – on zero results, systematically drops the most restrictive filter first

Introduced "Failure is Valid" messaging – the system never fabricates data; returns honest "no data found" responses with relaxation suggestions
Stack: Python 3.11+, TypeScript, FastMCP, FastAPI, Ollama, Sentence Transformers, React, SQL Server, Docker

OmniPilot-CX – AI Sales Agent for E-Commerce (Multi-Tenant SaaS)

Status: Production-Ready MVP

Jan 2026

Vertical SaaS transforming Moroccan e-commerce merchants' Facebook Messenger inquiries into 24/7 AI sales agents with category-specific personalities for Fashion, Beauty, and Electronics.

Built multi-tenant architecture supporting 1,000+ concurrent stores with sub-50ms product query responses from PostgreSQL cache, delivering 10-100x speed improvement over competitors using external APIs

Engineered in-app product management system with intelligent CSV/XLSX import featuring auto-column mapping for Arabic/French/English headers and category-specific field validation

Integrated Google Gemini + n8n workflows for multi-language conversation handling (Darija/French/English auto-detection), context-aware product recommendations, and automated order capture

Implemented platform-owned Facebook App with one-click OAuth onboarding, encrypted credential storage (AES-256-GCM), and proactive health monitoring system preventing silent token expiry failures

Tech: Next.js, Node.js, PostgreSQL, Prisma, n8n, Chatwoot, Google Gemini API, Facebook Graph API, Docker

Medical Appointment System – Full-Stack Healthcare SaaS

Status: Production-Ready

April 2025

Scalable medical scheduling platform connecting patients with healthcare providers through real-time appointment booking, doctor availability management, and secure role-based access control.

Architected monorepo structure using Turborepo with React.js frontend and Express.js backend, achieving 40% faster build times and streamlined dependency management

Implemented secure authentication system with JWT-based tokens and RBAC, reducing unauthorized access attempts to zero across 500+ test sessions

Developed RESTful API layer with MySQL and Sequelize ORM, including comprehensive validation using Zod schemas and automated error handling middleware

Built responsive patient interface using Shadcn UI and Tailwind CSS with appointment booking, doctor search, availability calendar, and real-time status updates via React Query

Tech: React, Express.js, Node.js, MySQL, Sequelize, JWT, Tailwind CSS, Shadcn UI, Docker, Turborepo

Enterprise LLM System – Offline AI/ML Integration

Confidential Client, Rabat

Jan 2025 – April 2025

Secure, offline AI system deployed on enterprise intranet for document analysis, policy compliance detection, and legal support using locally-hosted LLMs with zero external API dependencies.

Integrated open-source LLMs (Qwen, DeVstral, GLM, etc.) into secure offline environment using Ollama and Open WebUI, eliminating data privacy concerns

for confidential enterprise documents

Built custom RAG pipelines with n8n workflow orchestration and Qdrant VectorDB, processing documents with 95% accuracy in policy violation detection

Deployed specialized models via Modelfile for document summarization, compliance scoring, and context-aware legal recommendations tailored to company policies

Tech: Ollama, Open WebUI, n8n, Docker, Qdrant VectorDB, Prompt Engineering, RAG

AI Evaluation Scoring System – HR Performance Analytics

Confidential Client, Rabat

Feb 2024 – Jan 2025

Internal ML system analyzing narrative employee evaluations in French to generate quantitative performance scores, reducing HR review time by 60% while maintaining 92% accuracy against human assessments.

Fine-tuned CamemBERT model for French-language sentiment analysis and performance extraction from unstructured text evaluations, achieving 92% correlation with human-scored assessments

Deployed production API using FastAPI with <500ms inference latency, processing 200+ evaluations per hour during peak review cycles

Tech: Python, CamemBERT, FastAPI, Docker

Fuel Station Management System – MERN Stack + Computer Vision

Confidential Client, Rabat

Oct 2023 – Feb 2024

Backend system managing fuel transactions and station operations with automated Moroccan license plate recognition, reducing manual data entry errors by 85% and processing 500+ daily transactions.

Built RESTful backend using MERN stack with JWT-based authentication and role-based access for station managers, attendants, and administrators

Trained and deployed Detectron2 OCR model on custom dataset of Moroccan license plates, achieving 94% recognition accuracy in real-world lighting conditions

Containerized ML inference API using FastAPI and Docker, serving license plate scanning endpoints to mobile development team with 99.5% uptime

Tech: Node.js, Express.js, MongoDB, React.js, FastAPI, Docker, Detectron2, JWT

CORE EXPERTISE

- AI Agent Architecture: OpenClaw, multi-agent orchestration, autonomous agents, tool-using agents, agent routing
- Full-Stack Development: React, Next.js, Node.js, NestJS, Express.js, TypeScript, JavaScript (ES6+)
- AI & Machine Learning: Ollama, RAG, LangChain, n8n Workflow Automation, Prompt Engineering, Fine-Tuning, Google Gemini API, LLM-Oriented Architecture
- Web Scraping & Data Engineering: Firecrawl API, AI-driven extraction, ETL pipelines, data validation, deduplication, structured reporting
- SaaS & Multi-Tenant Architecture: PostgreSQL, Prisma ORM, Tenant Isolation, Encrypted Credential Management, Row-Level Security

AI Chatbot Development: Conversational AI, NLP, Multi-Language Support, Facebook Messenger Integration, Chatwoot, Customer Engagement Automation
API Development: REST APIs, JWT Authentication, OAuth 2.0, Secure Token Management, Rate Limiting, Webhook Integrations
DevOps & Deployment: Docker, Docker Compose, Git, GitHub, Vercel, NGINX, CI/CD Pipelines

TECHNICAL STACK

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Shadcn UI, Redux Toolkit, React Query, Vite

Backend: Node.js, Express.js, NestJS, FastAPI, ASP.NET MVC, Python, C#

Databases: PostgreSQL, MySQL, MongoDB, Qdrant VectorDB, Prisma, Sequelize, SQL Server

AI/ML: OpenCLAW, Ollama, Puter.js, n8n, Google Gemini, Claude, Open WebUI, LangChain, LangGraph, Agno, Firecrawl, CamemBERT, Detectron2, PyTorch, Transformers

DevOps: Docker, Git, GitHub, Vercel, NGINX, OpenTelemetry

EDUCATION

Diploma in Specialized Information Technology Development | 2020
Specialized Institute of Technology, Laayoune, Morocco

Undergraduate Degree in Mathematical Sciences | 2015
IBN BATOUTA High School, Laayoune, Morocco

LANGUAGES

Arabic: Native | English: Professional | French: Intermediate